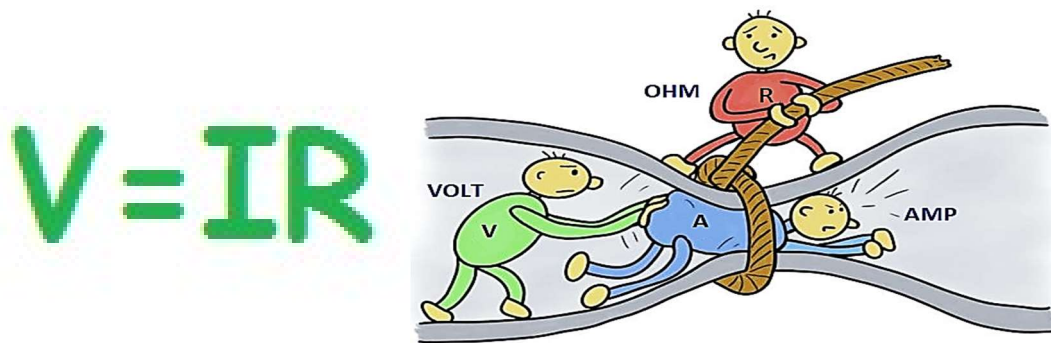


Ohm's Law

Ohm's law states the relationship between electric current and potential difference. The current that flows through most conductors is directly proportional to the voltage applied to it. Georg Simon Ohm, a German physicist was the first to verify Ohm's law experimentally. Ohm's Law is a formula used to calculate the relationship between voltage, current and resistance in an electrical circuit

Ohm's law states that the voltage across a conductor is directly proportional to the current flowing through it, provided all physical conditions and temperatures remain constant.

Mathematically, this current-voltage relationship is written as,



In the equation, the constant of proportionality, R , is called Resistance and has units of ohms, with the symbol Ω .

Where,

- R is the Constant of Proportionality known as Resistance,
- V is the Voltage applied, and
- I is the current flowing through the electrical circuit.

The same formula can be rewritten in order to calculate the current and resistance respectively as follows:

$$I = \frac{V}{R}$$

$$R = \frac{V}{I}$$

Ohm's law only holds true if the provided temperature and the other physical factors remain constant. In certain components, increasing the current raises the temperature. An example of this is the filament of a light bulb, in which the temperature rises as the current is increased. In this case, Ohm's law cannot be applied. **The lightbulb filament violates Ohm's Law.**

Ohm's Law Statement: Ohm's law states that the voltage across a conductor is directly proportional to the current flowing through it, provided all physical conditions and temperature, remain constant.

Ohm's Law Equation: $V = IR$, where V is the voltage across the conductor, I is the current flowing through the conductor and R is the resistance provided by the conductor to the flow of current.

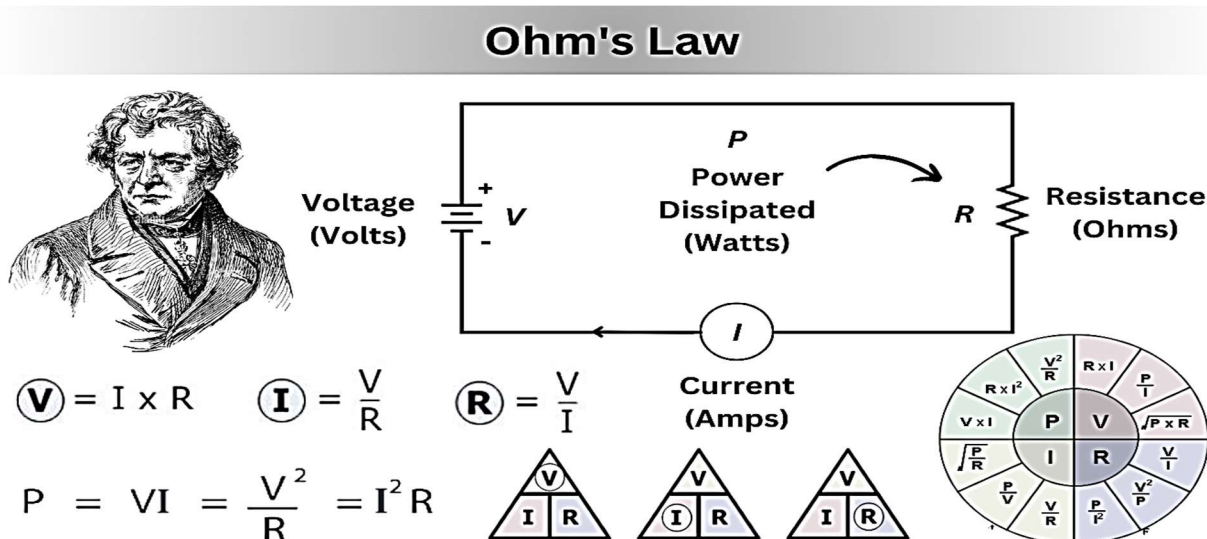
Experimental Verification of Ohm's Law

Ohm's Law can be easily verified by the following experiment:

Apparatus Required:

- Resistor
- Ammeter
- Voltmeter
- Battery
- Plug Key

Circuit Diagram:



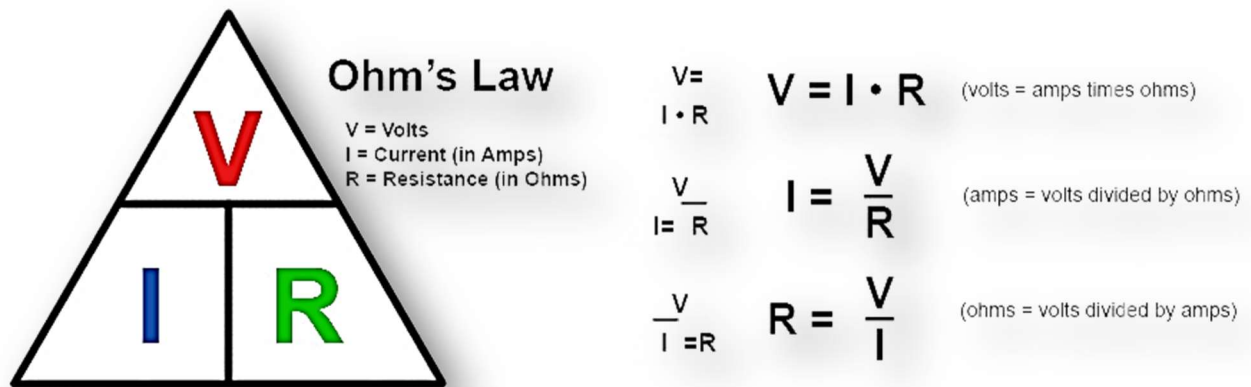
Procedure:

- Initially, the key K is closed and the rheostat is adjusted to get the minimum reading in ammeter A and voltmeter V .
- The current in the circuit is increased gradually by moving the sliding terminal of the resistance. During the process, the current flowing in the circuit and the corresponding value of potential difference across the resistance wire R are recorded.
- This way different sets of values of voltage and current are obtained.

- For each set of values of V and I, the ratio of V/I is calculated.
- When you calculate the ratio V/I for each case, you will come to notice that it is almost the same. So, $V/I = R$, which is a constant.
- Plot a graph of the current against the potential difference, it will be a straight line. This shows that the current is proportional to the potential difference.

Ohm's Law Magic Triangle

You can make use of Ohm's law magic triangle to remember the different equations for Ohm's law used to solve for different variables (V, I, R).



If the value of voltage is asked and the values of the current and resistance are given, then to calculate voltage simply cover V at the top. So, we are left with the I and R or $I \times R$. So, the equation for Voltage is Current multiplied by Resistance. Examples of how the magic triangle is employed to determine the voltage using Ohm's law are given below.

Example 1: If the resistance of an electric iron is 50Ω and a current of 3.2 A flows through the resistance. Find the voltage between two points.

Solution:

If we are asked to calculate the value of voltage with the value of current and resistance, then cover V in the triangle. Now, we are left with I and R or more precisely $I \times R$.

Therefore, we use the following formula to calculate the value of V:

$$V = I \times R$$

Substituting the values in the equation, we get

$$V = 3.2 \text{ A} \times 50 \Omega = 160 \text{ V}$$

Calculating Electrical Power Using Ohm's Law

The rate at which energy is converted from the electrical energy of the moving charges to some other form of energy like mechanical energy, heat energy, energy stored in magnetic fields or electric fields, is known as electric power. The unit of power is the watt. The electrical power can be calculated using Ohm's law and by substituting the values of voltage, current and resistance.

Formula to find power

When the values for voltage and current are given,

$$P = VI$$

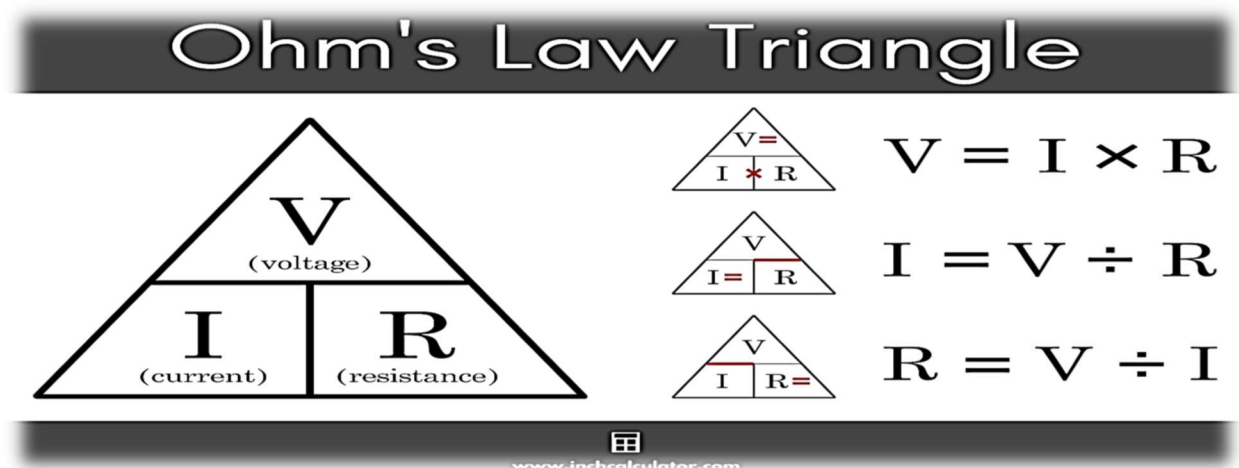
When the values for voltage and resistance are given,

$$P = V^2 \div R$$

When the values for current and resistance are given,

$$P = I^2 R$$

What is a Power Triangle?



The power triangle can be employed to determine the value of electric power, voltage and current when the values of the other two parameters are given to us. In the power triangle, the power (P) is on the top and current (I) and voltage (V) are at the bottom.

When the values of current and voltage are given, the formula for finding power is,

$$P = VI$$

When the values of power and voltage are given, the formula for finding current is,

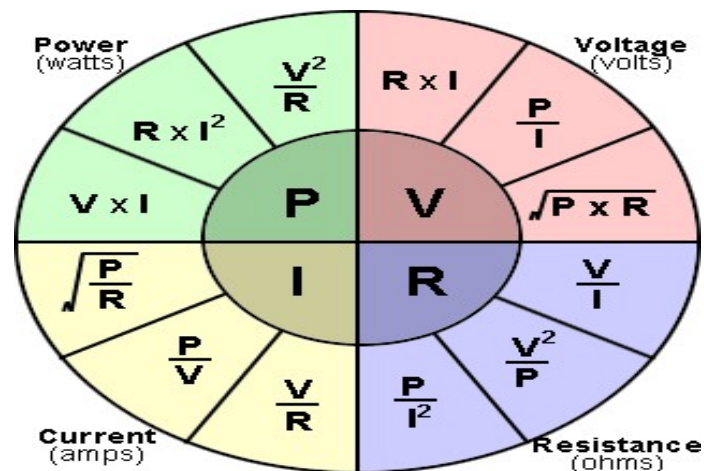
$$I = P/V$$

When the values of power and current are given, the formula for finding voltage is,

$$V = P/I$$

Ohm's Law Pie Chart

To better understand the relationship between various parameters, we can take all the equations used to find the voltage, current, resistance and power, and condense them into a simple Ohm's Law pie chart as shown below.



Ohm's Law Matrix Table

Like Ohm's Law Pie Chart shown above, we can condense the individual Ohm's Law equations into a simple matrix table as shown below for easy reference when calculating an unknown value.

Ohm's law formulas		To Calculate			
		Voltage (V)	Current (I)	Resistance (R)	Power (P)
Given parameters	Power & Resistance	$V = \sqrt{P \cdot R}$	$I = \sqrt{P/R}$	---	---
	Voltage & Power	---	$I = \frac{P}{V}$	$R = \frac{V^2}{P}$	---
	Voltage & Resistance	---	$I = \frac{V}{R}$		$P = \frac{V^2}{R}$
	Voltage & Current	---	---	$R = \frac{V}{I}$	$P = VI$
	Current & Power	$V = \frac{P}{I}$	---	$R = \frac{P}{I^2}$	---
	Current & Resistance	$V = I \cdot R$	---	---	$P = I^2 R$

Ohm's Law Applications

The main applications of Ohm's law are:

- To determine the voltage, resistance or current of an electric circuit.
- Ohm's law maintains the desired voltage drop across the electronic components.
- Ohm's law is also used in DC ammeter and other DC shunts to divert the current.
- It also simplifies power calculations.
- To keep the desired voltage, drop between the electrical components, Ohm's law is employed.
- An electric circuit's voltage, resistance, or current must be determined.
- Ohm's law is also utilized to redirect current in DC ammeters and other DC shunts.

Limitations of Ohm's Law

Following are the limitations of Ohm's law:

- Ohm's law is not applicable for unilateral electrical elements like diodes and transistors as they allow the current to flow through in one direction only.
- For non-linear electrical elements with parameters like capacitance, resistance etc the ratio of voltage and current won't be constant with respect to time making it difficult to use Ohm's law.

Kirchhoffs First Law – The Current Law, (KCL)

Kirchhoffs Current Law or KCL, states that *“The total current or charge entering a junction or node is precisely equal to the total current or charge exiting the node, as no charge is lost at the node”*.

To put it another way, the algebraic sum of all currents entering and exiting a node must be zero. Kirchhoff's Current Law (KCL) is also known as Kirchhoff's First Law or Kirchhoff's Junction Rule.

The algebraic sum of currents entering any node in a circuit is zero.

Kirchhoffs Current Law

Kirchhoff's First law is similar to the **Law of Conservation of charge**. As a result, a Node or junction is a point in a circuit that does not serve as a charge source or sink. Therefore,

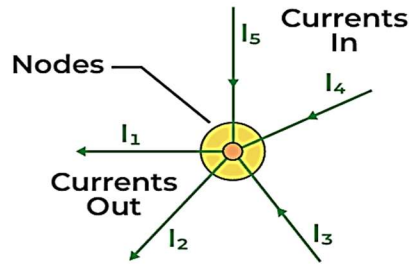
$$\sum_{k=1}^n I = 0$$

Where n denotes the total number of branches at the node with currents flowing toward or away from it.

$$I_{(\text{exiting})} + I_{(\text{entering})} = 0$$

Currents Entering the Node Equals Current Leaving the Node

$$-I_1 - I_2 + I_3 + I_4 + I_5 = 0$$



In the figure shown below the node or the junction has five branches. The three incoming currents, i_3 , i_4 , and i_5 , and the two outgoing currents, i_1 and i_2 . Hence, according to Kirchhoff's Current Law, the sum of total incoming and outgoing currents at the node will now equal zero. Consider the current entering the node as positive and the current exiting the node as negative then the algebraic sum can be represented as

$$(-I_1) + (-I_2) + (I_3) + (I_4) + (I_5) = 0$$

$$\Rightarrow I_3 + I_4 + I_5 = I_1 + I_2$$

It should be noted that there are no set rules for assigning positive and negative signs to the current. However, to avoid confusion entering current is taken as positive, and exiting current is taken as negative.

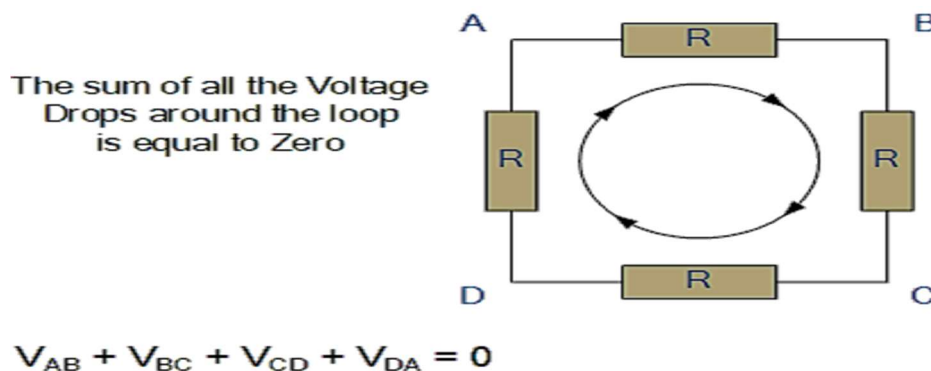
Kirchhoffs Second Law – The Voltage Law, (KVL)

Kirchhoffs Voltage Law or KVL, states that “*The voltage drop around a loop equals the algebraic sum total of the voltage drop across every electrical component linked in the same loop for any closed network and is equal to zero*”. In other words, the algebraic sum of all voltage sources and voltage drops within a closed loop must be equal to zero since the algebraic sum of the voltage drops equals the algebraic sum of the voltage sources. This idea by Kirchhoff is known as the **Conservation of Energy**.

The algebraic sum of the voltages around any loop is zero.

Kirchhoffs Voltage Law

Kirchhoff's Second Law is similar to the Law of Conservation of energy. Because the net change in the energy of a charge after it completes a closed route must be zero.



So, for a closed loop,

$$\sum_{k=1}^n V_k = 0$$

The total number of electrical components in the loop is given by n.

$$V_{PQ} + V_{QR} + V_{RS} + V_{SP} = 0$$

Starting at any point in the loop continue in the **same direction** noting the direction of all the voltage drops, either positive or negative, and returning back to the same starting point. It is important to maintain the same direction either clockwise or anti-clockwise or the final voltage sum will not be equal to zero. We can use Kirchhoff's voltage law when analyzing series circuits.

When analyzing either DC circuits or AC circuits using **Kirchhoffs Circuit Laws** a number of definitions and terminologies are used to describe the parts of the circuit being analyzed such as: node, paths, branches, loops and meshes. These terms are used frequently in circuit analysis so it is important to understand them.

Sign Conventions for Kirchhoff's Law

Use the following convention to assign sign in the circuit provided

- Take Positive sign for current entering the junction and negative for current leaving the junction
- In a loop, a rise in potential difference or EMF from lower to higher is always seen as positive i.e. moving from negative to positive terminal
- In a loop, a reduction in potential difference or EMF from higher to lower is always seen as negative i.e. moving from positive to negative terminal.
- If the looping direction is the same as the current flowing through the circuit, the voltage drops across the resistor is considered negative.

Uses of Kirchhoff's Law

Kirchhoff's Law is used for following

- It is used to find out how much current is flowing and how much voltage is dropped in various areas of the complicated circuit.
- It aids in determining the direction of current in various circuit loops.
- Kirchhoff's Laws can help you comprehend how energy moves via an electric circuit.

Limitations of Kirchhoff's Laws

There are various limitations associated with Kirchhoff's Laws which are:

- Kirchhoff's Law has one fundamental flaw: it implies that there is no changing magnetic field throughout the loop's region, which might induce a change in magnetic flux and the creation of EMF in the circuit. This might result in a calculation mistake for high-frequency AC circuits.
- Kirchhoff also ignored the impact of the electric field generated by other circuit components.

- Kirchhoff's Current Law works with the premise that current only travels through conductors and wires. While parasitic capacitance can no longer be overlooked in High-Frequency circuits. Because conductors or wires are serving as transmission lines in certain circumstances, electricity may start flowing in an open circuit.

P-N Junction

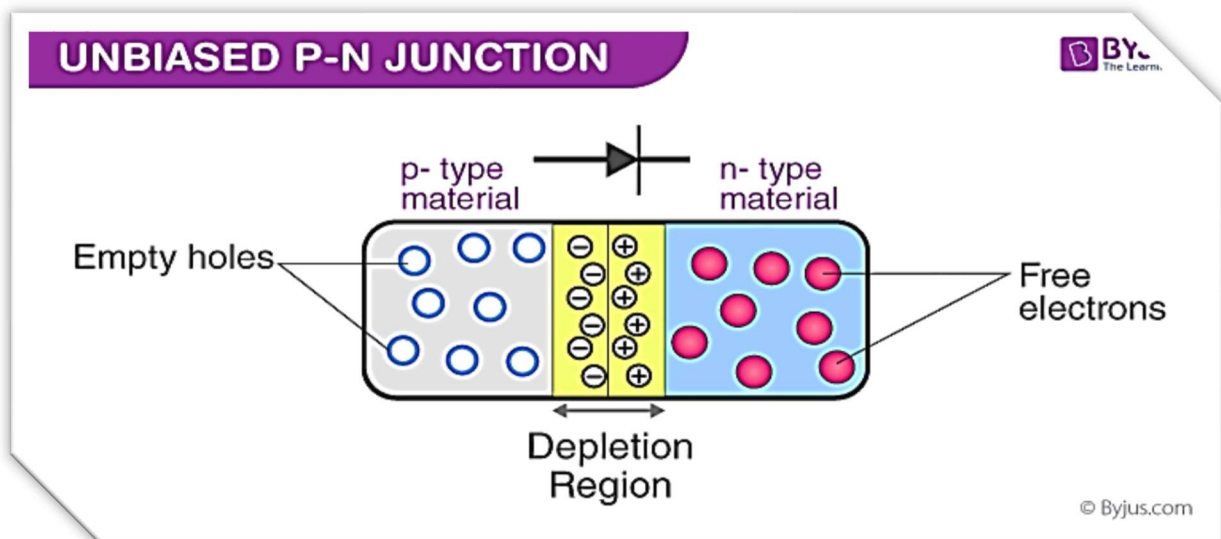
What is P-N Junction?

A P-N junction is an interface or a boundary between two semiconductor material types, namely the p-type and the n-type, inside a semiconductor.

In a semiconductor, the P-N junction is created by the method of doping. The p-side or the positive side of the semiconductor has an excess of holes, and the n-side or the negative side has an excess of electrons. The process of doping is explained in further detail in the next section.

Formation of P-N Junction

As we know, if we use different semiconductor materials to make a P-N junction, there will be a grain boundary that would inhibit the movement of electrons from one side to the other by scattering the electrons and holes and thus, we use the process of doping. We will understand the process of doping with the help of this example. Let us consider a thin p-type silicon semiconductor sheet. If we add a small amount of pentavalent impurity to this, a part of the p-type Si will get converted to n-type silicon. This sheet will now contain both the p-type region and the n-type region and a junction between these two regions. The processes that follow after forming a P-N junction are of two types – diffusion and drift. There is a difference in the concentration of holes and electrons at the two sides of a junction. The holes from the p-side diffuse to the n-side, and the electrons from the n-side diffuse to the p-side. These give rise to a diffusion current across the junction.



Also, when an electron diffuses from the n-side to the p-side, an ionized donor is left behind on the n-side, which is immobile. As the process goes on, a layer of positive charge is developed on the n-side of the junction. Similarly, when a hole goes from the p-side to the n-side, an ionized acceptor is left behind on the p-side, resulting in the formation of a layer of negative charges in the p-side of the junction. This region of positive charge and negative charge on either side of the junction is termed as the depletion region. Due to this positive space charge region on either side of the junction, an electric field with the direction from a positive charge towards the negative charge is developed. Due to this electric field, an electron on the p-side of the junction moves to the n-side of the junction. This motion is termed the drift. Here, we see that the direction of the drift current is opposite to that of the diffusion current.

Biasing Conditions for the P-N Junction Diode

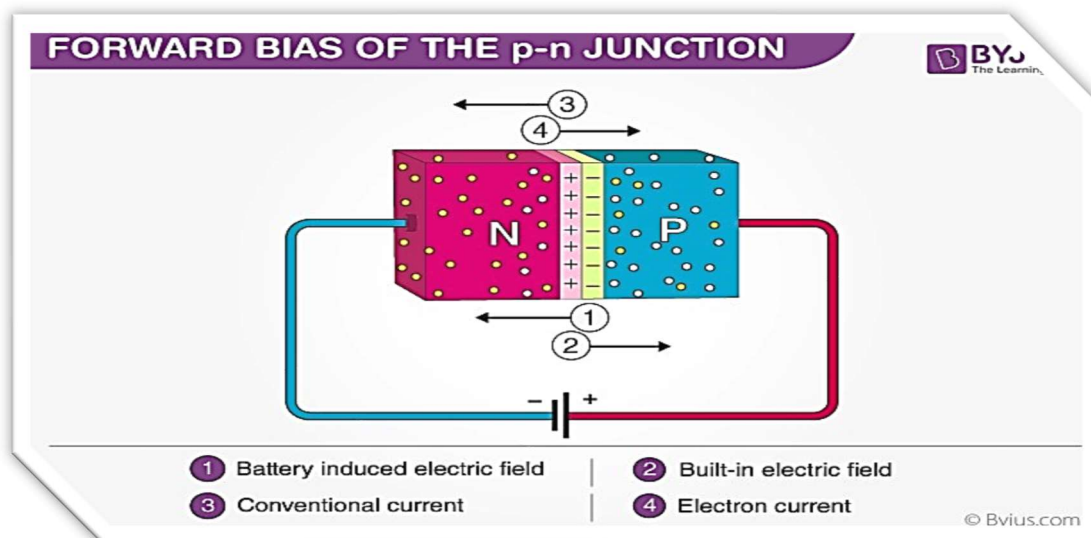
There are two operating regions in the P-N junction diode:

- P-type
- N-type

There are three biasing conditions for the P-N junction diode, and this is based on the voltage applied:

- Zero bias: No external voltage is applied to the P-N junction diode.
- Forward bias: The positive terminal of the voltage potential is connected to the p-type while the negative terminal is connected to the n-type.
- Reverse bias: The negative terminal of the voltage potential is connected to the p-type and the positive is connected to the n-type.

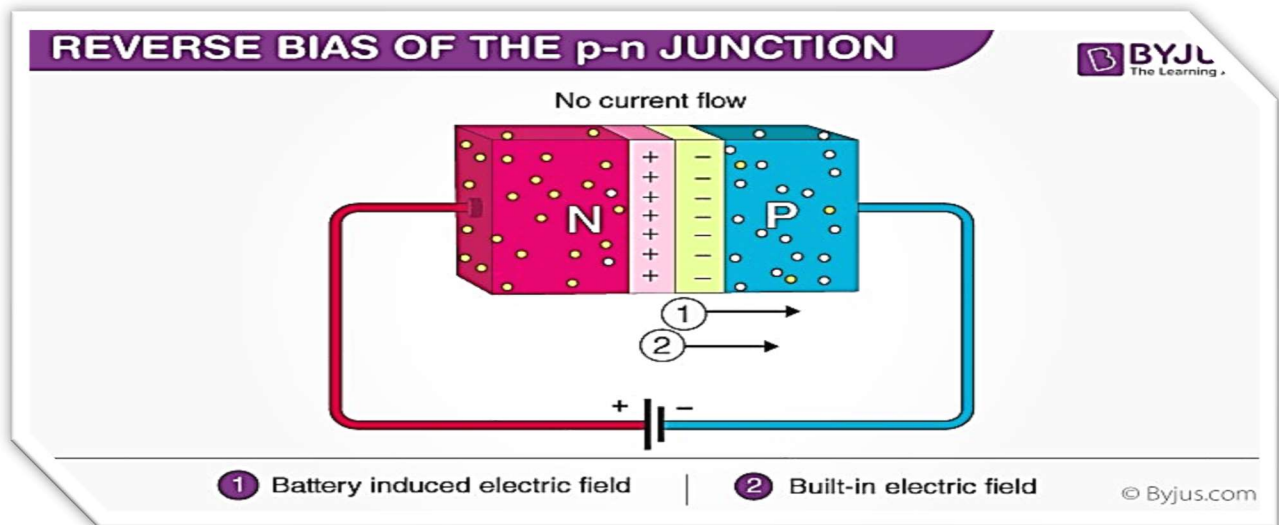
Forward Bias



When the p-type is connected to the battery's positive terminal and the n-type to the negative terminal, then the P-N junction is said to be forward-biased. When the P-N junction is forward biased, the built-in electric field at the P-N junction and the applied electric field are in opposite directions. When both the electric fields add up, the resultant electric field has a magnitude lesser

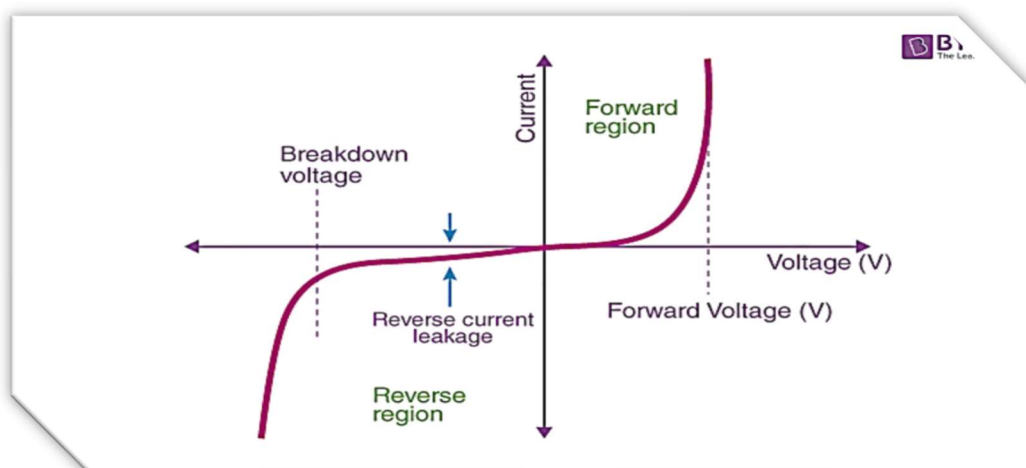
than the built-in electric field. This results in a less resistive and thinner depletion region. The depletion region's resistance becomes negligible when the applied voltage is large. In silicon, at the voltage of 0.6 V, the resistance of the depletion region becomes completely negligible, and the current flows across it unimpeded.

Reverse Bias



When the p-type is connected to the battery's negative terminal and the n-type is connected to the positive side, the P-N junction is reverse biased. In this case, the built-in electric field and the applied electric field are in the same direction. When the two fields are added, the resultant electric field is in the same direction as the built-in electric field, creating a more resistive, thicker depletion region. The depletion region becomes more resistive and thicker if the applied voltage becomes larger.

V-I Characteristics of P-N Junction Diode



VI characteristics of P-N junction diodes is a curve between the voltage and current through the circuit. Voltage is taken along the x-axis while the current is taken along the y-axis. The above graph is the V-I characteristics curve of the P-N junction diode. With the help of the curve, we can understand that there are three regions in which the diode works, and they are:

- Zero bias
- Forward bias
- Reverse bias

When the P-N junction diode is in zero bias condition, there is no external voltage applied and this means that the potential barrier at the junction does not allow the flow of current.

When the P-N junction diode is in forward bias condition, the p-type is connected to the positive terminal while the n-type is connected to the negative terminal of the external voltage. When the diode is arranged in this manner, there is a reduction in the potential barrier. For silicone diodes, when the voltage is 0.7 V and for germanium diodes, when the voltage is 0.3 V, the potential barriers decrease, and there is a flow of current.

When the diode is in forward bias, the current increases slowly, and the curve obtained is non-linear as the voltage applied to the diode overcomes the potential barrier. Once the diode overcomes the potential barrier, the diode behaves normally, and the curve rises sharply as the external voltage increases, and the curve obtained is linear.

When the P-N junction diode is in negative bias condition, the p-type is connected to the negative terminal while the n-type is connected to the positive terminal of the external voltage. This results in an increase in the potential barrier. Reverse saturation current flows in the beginning as minority carriers are present in the junction.

When the applied voltage is increased, the minority charges will have increased kinetic energy which affects the majority charges. This is the stage when the diode breaks down. This may also destroy the diode.

Applications of P-N Junction Diode

- P-N junction diode can be used as a photodiode as the diode is sensitive to the light when the configuration of the diode is reverse-biased.
- It can be used as a solar cell.
- When the diode is forward-biased, it can be used in LED lighting applications.
- It is used as rectifier in many electric circuits and as a voltage-controlled oscillator in varactors.

Zener Diode

A Zener Diode, also referred to as a breakdown diode, is a specially doped semiconductor device engineered to function in the reverse direction. When the voltage across a Zener diode's terminals is reversed and reaches the Zener Voltage (also known as the knee voltage), the junction experiences a breakdown, allowing current to flow in the opposite direction. This phenomenon, known as the Zener Effect, is a key characteristic of Zener diodes.

A Zener diode is a highly doped semiconductor device specifically designed to function in the reverse direction. It is engineered with a wide range of Zener voltages (V_Z), and certain types are even adjustable to achieve variable voltage regulation.

How does a Zener Diode work in reverse bias?

A Zener diode functions similarly to a regular diode when forward-biased. However, in reverse-biased mode, a small leakage current flows through the diode. As the reverse voltage increases and reaches the predetermined breakdown voltage (V_Z), current begins to flow through the diode. This current reaches a maximum level determined by the series resistor, after which it stabilizes and remains constant across a wide range of applied voltages.

There are two types of breakdowns in a Zener Diode: **Avalanche Breakdown and Zener Breakdown.**

Avalanche Breakdown in Zener Diode

Avalanche breakdown occurs in both normal diodes and Zener diodes when subjected to high reverse voltage. When a significant reverse voltage is applied to the PN junction, the free electrons gain enough energy to accelerate at high velocities. These high-velocity electrons collide with other atoms, causing the ejection of additional electrons. This continuous collision process generates a large number of free electrons, resulting in a rapid increase in electric current through the diode. In the case of a normal diode, this sudden surge in current could permanently damage it. However, a Zener diode is specifically designed to withstand avalanche breakdown and can handle the sudden current spike. Avalanche breakdown typically occurs in Zener diodes with a Zener voltage (V_Z) greater than 6V.

Zener Breakdown in Zener Diode

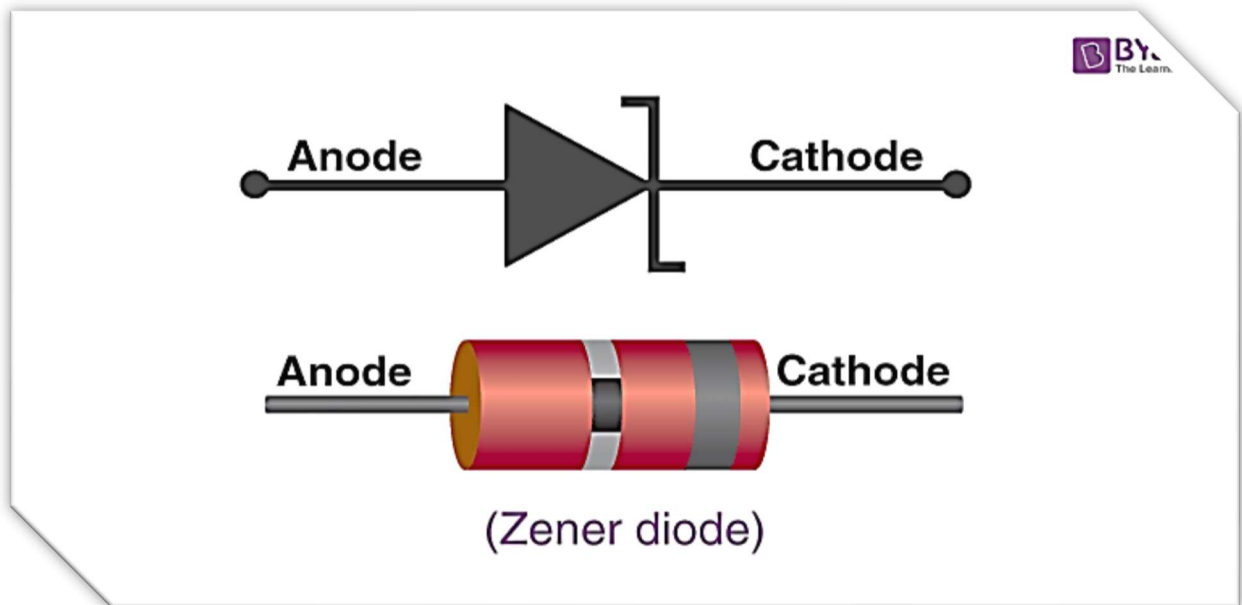
When the reverse bias voltage applied to a Zener diode approaches its Zener voltage, the electric field within the depletion region becomes strong enough to attract and remove electrons from their valence band. These valence electrons, energized by the intense electric field, break free from their parent atoms. This phenomenon takes place in the Zener breakdown region, where even a slight increase in voltage leads to a rapid surge in electric current.

Avalanche Breakdown vs. Zener Breakdown

The Zener effect is predominant in voltages up to 5.6 volts, while the avalanche effect becomes more prominent beyond that threshold. Although both effects are similar, the distinction lies in the fact that the Zener effect is a quantum phenomenon, whereas the avalanche effect involves the movement of electrons in the valence band, similar to an electric current. The avalanche effect allows a larger current through the diode compared to what a Zener breakdown would permit.

Circuit Symbol of Zener Diode

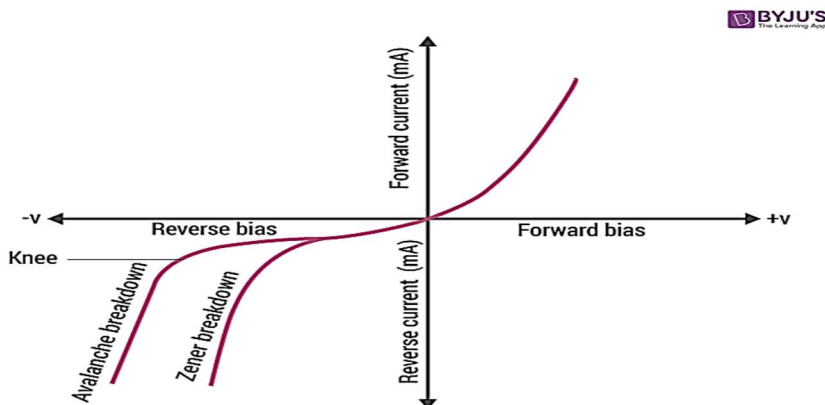
Zener diodes come in various packaging options, depending on their power dissipation requirements. Some are designed for high-power applications, while others are available in surface mount formats. The most commonly used Zener diode is packaged in a small glass enclosure, with a distinctive band indicating the cathode side of the diode.



The symbol used to represent a Zener diode in circuit diagrams is similar to that of a regular diode, but with a unique addition. It consists of a triangle or arrowhead pointing towards the cathode side (the side with the band) of the diode. This triangle is accompanied by two perpendicular lines at the cathode end, one extending upwards and the other extending downwards. These lines indicate the specific behavior of the Zener diode and help distinguish it from other types of diodes in circuit diagrams. The symbol provides a visual representation that allows engineers and technicians to easily identify and understand the presence of a Zener diode in a circuit.

V-I Characteristics of Zener Diode

The diagram given below shows the V-I characteristics of the Zener diode.



When reverse-biased voltage is applied to a Zener diode, it allows only a small amount of leakage current until the voltage is less than Zener voltage.

The V-I characteristics of a Zener diode can be divided into two parts as follows:

- **Forward Characteristics.**
- **Reverse Characteristics.**

Forward Characteristics of Zener Diode

The first quadrant in the graph represents the forward characteristics of a Zener diode. From the graph, we understand that it is almost identical to the forward characteristics of P-N junction diode.

Reverse Characteristics of Zener Diode

When a reverse voltage is applied to a Zener voltage, a small reverse saturation current I_o flows across the diode. This current is due to thermally generated minority carriers. As the reverse voltage increases, at a certain value of reverse voltage, the reverse current increases drastically and sharply. This is an indication that the breakdown has occurred. We call this voltage breakdown voltage or Zener voltage, and V_z denotes it.

Zener Diode Specifications

Some commonly used specifications for Zener diodes are as follows:

- **Zener/Breakdown Voltage** – The Zener or the reverse breakdown voltage ranges from 2.4 V to 200 V, sometimes it can go up to 1 kV while the maximum for the surface-mounted device is 47 V.
- **Current I_z (max)** – It is the maximum current at the rated Zener Voltage ($V_z - 200\mu\text{A}$ to 200 A)
- **Current I_z (min)** – It is the minimum value of current required for the diode to break down.

- **Power Rating** – It denotes the maximum power the Zener diode can dissipate. It is given by the product of the voltage of the diode and the current flowing through it.
- **Temperature Stability** – Diodes around 5 V have the best stability
- **Voltage Tolerance** – It is typically $\pm 5\%$
- **Zener Resistance (R_z)** – It is the resistance to the Zener diode exhibits.

Application of Zener Diode

Following are the applications of Zener diode:

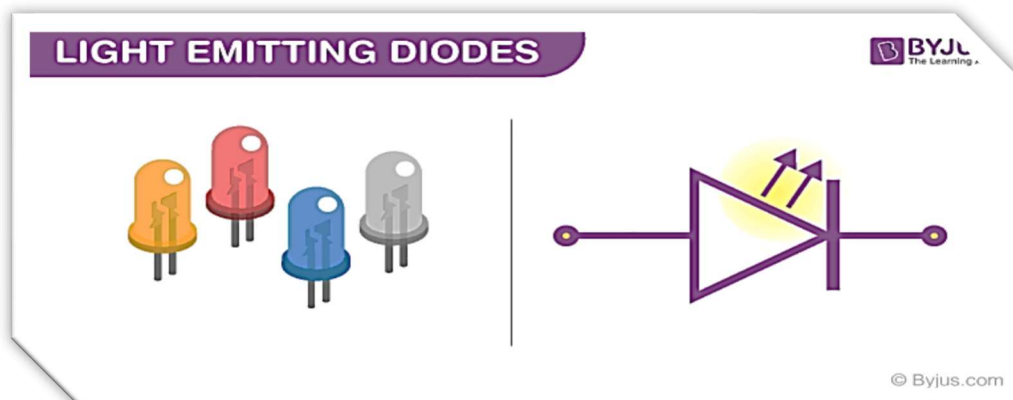
- **Zener diode as a voltage regulator:**
The Zener diode is used as a Shunt voltage regulator for regulating voltage across small loads. The Zener diode is connected parallel to the load to make it reverse bias, and once the Zener diode exceeds knee voltage, the voltage across the load will become constant. The breakdown voltage of Zener diodes will be constant for a wide range of currents.
- **Zener diode in over-voltage protection:**
When the input voltage is higher than the Zener breakage voltage, the voltage across the resistor drops resulting in a short circuit, this can be avoided by using the Zener diode.
- **Zener diode in clipping circuits:**
Zener diode is used for modifying AC waveform clipping circuits by limiting the parts of either one or both the half cycles of an AC waveform.

Light Emitting Diode (LED)

Light-emitting diode (LED) is a widely used standard source of light in electrical equipment. It has a wide range of applications ranging from your mobile phone to large advertising billboards. They mostly find applications in devices that show the time and display different types of data.

What is LED?

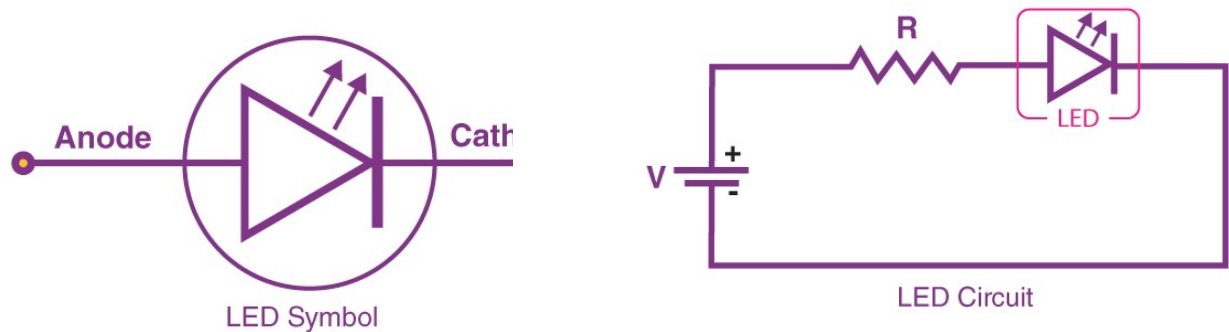
A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it. When current passes through an LED, the electrons recombine with holes emitting light in the process. LEDs allow the current to flow in the forward direction and blocks the current in the reverse direction.



Light-emitting diodes are heavily doped p-n junctions. Based on the semiconductor material used and the amount of doping, an LED will emit colored light at a particular spectral wavelength when forward biased. As shown in the figure, an LED is encapsulated with a transparent cover so that emitted light can come out.

LED Symbol

The LED symbol is the standard symbol for a diode, with the addition of two small arrows denoting the emission of light.

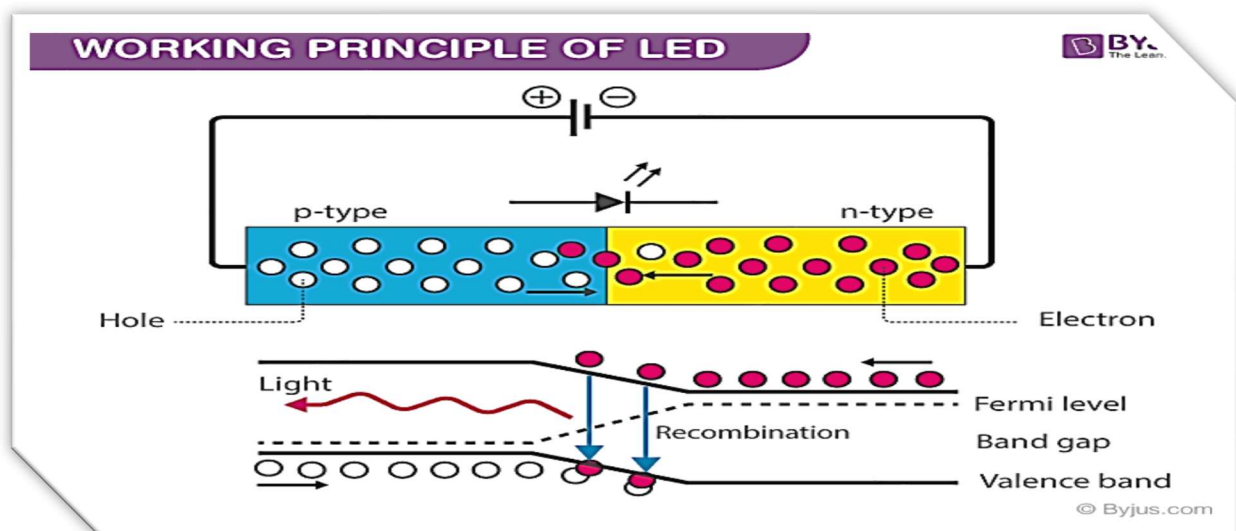


Simple LED Circuit

The figure below shows a simple LED circuit. The circuit consists of an LED, a voltage supply and a resistor to regulate the current and voltage.

How does an LED work?

When the diode is forward biased, the minority electrons are sent from $p \rightarrow n$ while the minority holes are sent from $n \rightarrow p$. At the junction boundary, the concentration of minority carriers increases. The excess minority carriers at the junction recombine with the majority charges carriers.



The energy is released in the form of photons on recombination. In standard diodes, the energy is released in the form of heat. But in light-emitting diodes, the energy is released in the form of photons. We call this phenomenon electroluminescence. Electroluminescence is an optical phenomenon, and electrical phenomenon where a material emits light in response to an electric current passed through it. As the forward voltage increases, the intensity of the light increases and reaches a maximum.

What determines the color of an LED?

The colour of an LED is determined by the material used in the semiconducting element. The two primary materials used in LEDs are aluminium gallium indium phosphide alloys and indium gallium nitride alloys. Aluminium alloys are used to obtain red, orange and yellow light, and indium alloys are used to get green, blue and white light. Slight changes in the composition of these alloys change the colour of the emitted light.

The major uses of LED (Light Emitting Diodes) are to illuminate objects and even places. Its application is everywhere due to its compact size, low consumption of energy, extended lifetime, and flexibility in terms of use in various applications.

Applications and uses of LEDs can be seen in:

- TV Backlighting
- Smartphone Backlighting
- LED displays
- Automotive Lighting
- Dimming of lights

Rectifier

Many electronic circuits use DC voltage for operation. We can easily convert AC voltage or current into DC voltage or current by using a device known as a p-n junction diode. A p-n junction diode allows electric current to flow in forward bias condition and blocks the current in reverse bias condition. Simply, a diode allows electric current to flow in one direction only. This unique property of diode allows it to act as a rectifier. Let us learn more about rectifiers in this article.

What is Rectifier?

A rectifier is an electronic device that converts an alternating current into a direct current by using one or more P-N junction diodes. A diode behaves as a one-way valve that allows current to flow in a single direction. This process is known as rectification.

A rectifier can take the shape of several physical forms such as solid-state diodes, vacuum tube diodes, mercury-arc valves, silicon-controlled rectifiers, and various other silicon-based semiconductors switches.

Different Types of Rectifiers

Rectifiers are mainly classified into two types as:

- Uncontrolled Rectifier
- Controlled Rectifier

Uncontrolled Rectifiers

The type of rectifier whose voltage cannot be controlled is known as an uncontrolled rectifier. Uncontrolled rectifiers are further divided as follows:

- Half Wave Rectifier
- Full Wave Rectifier

The type of rectifier that converts only the half cycle of the alternating current into the direct current is known as a half-wave rectifier. Likewise, a full-wave rectifier converts both positive and negative half cycles of the AC. An example of this is a bridge rectifier. A bridge rectifier uses 4 diodes that are connected in the form of a Wheatstone bridge.

Controlled Rectifiers

A type of rectifier whose voltage can be varied is known as the controlled rectifier. We use SCRs, MOSFETs and IGBTs to make an uncontrolled rectifier a controlled one. These rectifiers are preferred over their uncontrolled counterparts. There are two types of controlled rectifiers, and they are Half Wave Controlled Rectifier and Full Wave Controlled Rectifier. Half-wave controlled rectifier has the same design as the half-wave uncontrolled rectifier except we replace the diode with an SCR.

Advantages of Rectifiers

Main advantages of rectifiers are:

- Converts AC to DC power for electronic devices.
- Essential for battery charging in devices and electric vehicles.
- Used in industrial processes like electroplating and welding.
- Enables efficient HVDC power transmission over long distances.
- Provides controlled and stable DC power for precision applications.
- Integral in renewable energy systems for converting variable AC to DC.
- Facilitates signal processing by converting AC signals to rectified signals.
- Contributes to energy efficiency in power conversion processes.
- Cost-effective and reliable for a wide range of applications.

Disadvantages of Rectifiers

Key disadvantages of rectifiers are:

- Generates harmonics and electrical noise during the conversion process.
- May require additional filtering components to reduce output ripple.
- Limited in handling high power levels compared to some alternative technologies.
- Efficiency can decrease at partial loads or when operating away from design conditions.
- High initial costs for certain types of rectifiers and associated equipment.
- Some rectifiers may have a relatively lower power factor, leading to poor power quality.
- Maintenance may be required, especially in heavy industrial applications.
- Possibility of heat generation, requiring cooling mechanisms in high-power rectifiers.
- Incompatible with AC-powered devices, necessitating additional conversion for certain applications.
- Environmental concerns, as certain types of rectifiers may contain hazardous materials.

Applications of Rectifiers

Some of the key applications of rectifiers are:

- It is used for **powering appliances** as we know all electrical appliances use a DC power supply to function. By using rectifier in power supply and systems it helps in conversion of AC to DC.
- It is used to soldering and modulation as it is used in mosquito repellent to power the lead for the fumes.
- **Voltage multiplier** - As said earlier that rectifiers are used for power conversion such as AC power to DC power and so on same as they are used to convert AC to DC voltage for electronic appliances such as microwave ovens and so on.
- Used in **Transformers** to obtain desired voltage by using rectifier in combination with a step-down or step-up transformer.
- For **Radio appliances** rectifiers acts as an detector because the output consists of an audio signals. For examples Modulation, Demodulation and power supplies.

Half Wave Rectifier

Half-wave rectifiers transform AC voltage to DC voltage. A halfwave rectifier circuit uses only one diode for the transformation. A halfwave rectifier is defined as a type of rectifier that allows only one-half cycle of an AC voltage waveform to pass while blocking the other half cycle. In this session, let us know in detail about the half-wave rectifier.

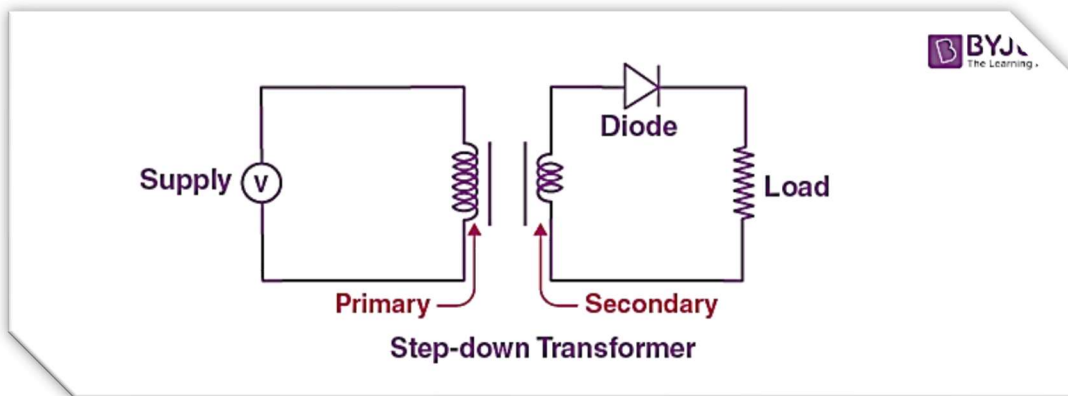
Half Wave Rectifier Circuit

A half-wave rectifier is the simplest form of the rectifier and requires only one diode for the construction of a halfwave rectifier circuit.

A halfwave rectifier circuit consists of three main components as follows:

- A diode
- A transformer
- A resistive load

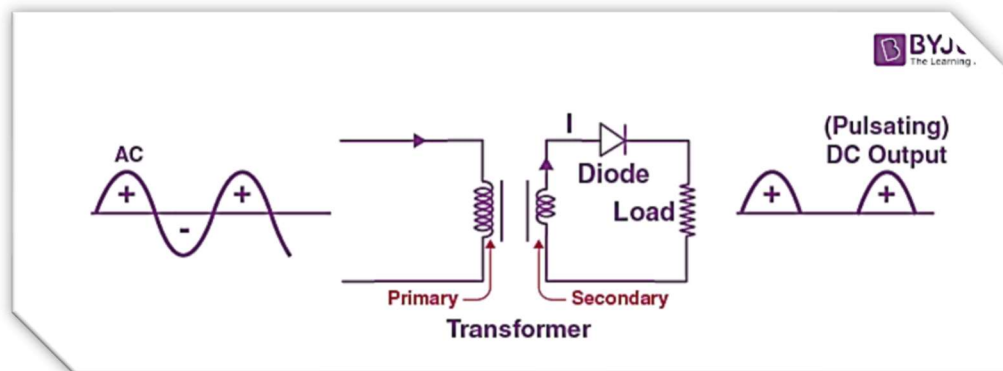
Given below is the half-wave rectifier diagram:



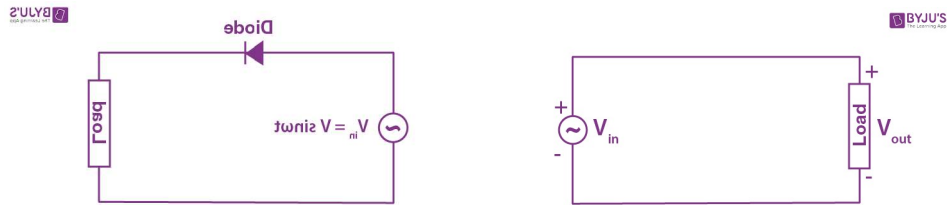
Working of Half Wave Rectifier

In this section, let us understand how a half-wave rectifier transforms AC into DC.

- A high AC voltage is applied to the primary side of the step-down transformer. The obtained secondary low voltage is applied to the diode.
- The diode is forward biased during the positive half cycle of the AC voltage and reverse biased during the negative half cycle.
- The final output voltage waveform is as shown in the figure below:

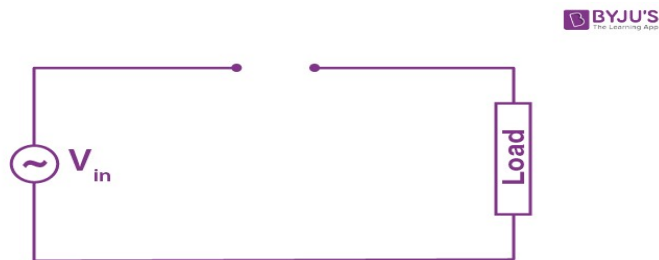


For better understanding, let us simplify the half-wave circuit by replacing the secondary transformer coils with a voltage source as shown below:



For the positive half cycle of the AC source voltage, the circuit effectively becomes as shown below in the diagram:

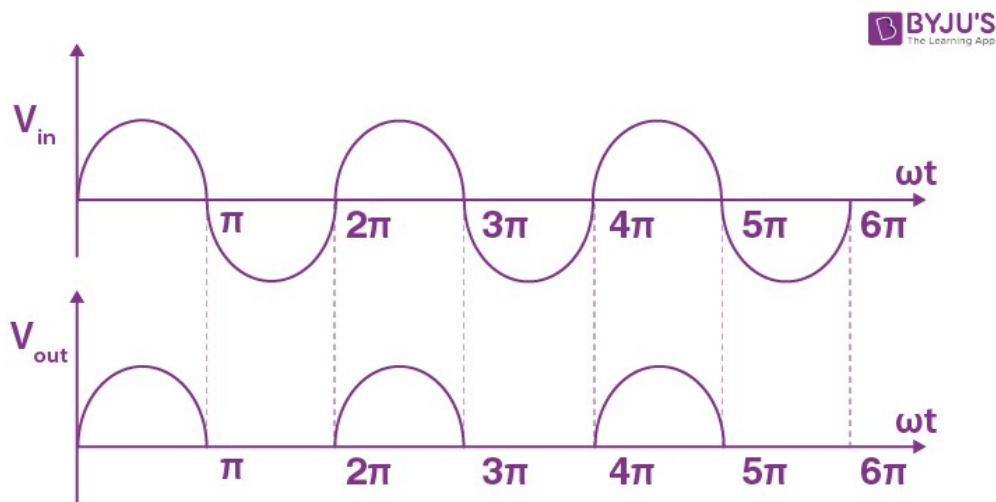
When the diode is forward biased, it acts as a closed switch. But, during the negative half cycle of the AC source voltage, the equivalent circuit becomes as shown in the figure below



When a diode is reverse biased, it acts as an open switch. Since no current can flow to the load, the output voltage is equal to zero.

Half Wave Rectifier Waveform

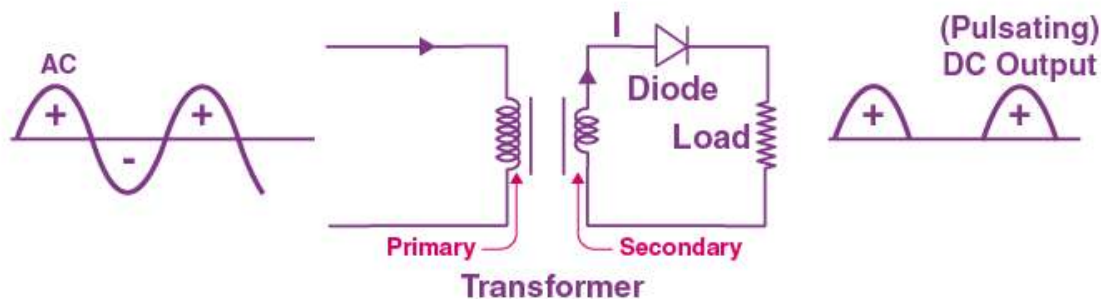
The halfwave rectifier waveform before and after rectification is shown below in the figure.



Half Wave Rectifier Capacitor Filter

The output waveform of a halfwave rectifier is a pulsating DC waveform. Filters in halfwave rectifiers are used to transform the pulsating waveform into constant DC waveforms. A capacitor or an inductor can be used as a filter.

The circuit diagram below shows how a capacitive filter is used with halfwave rectifier to smoothen out a pulsating DC waveform into a constant DC waveform.



Applications of Half Wave Rectifier

Here are a few common applications of half wave rectifiers:

- They are used for signal demodulation purpose
- They are used for rectification applications
- They are used for signal peak applications

Disadvantages of Half Wave Rectifier

- Power loss
- Low output voltage
- The output contains a lot of ripples

Full Wave Rectifier

Electric circuits that convert AC to DC are known as rectifiers. Rectifiers are classified into two types as Half Wave Rectifiers and Full Wave Rectifiers. Significant power is lost while using a half-wave rectifier and is not feasible for applications that need a smooth and steady supply. For a more smooth and steady supply, we use the full wave rectifiers. In this article, we will be looking into the working and characteristics of a full wave rectifier.

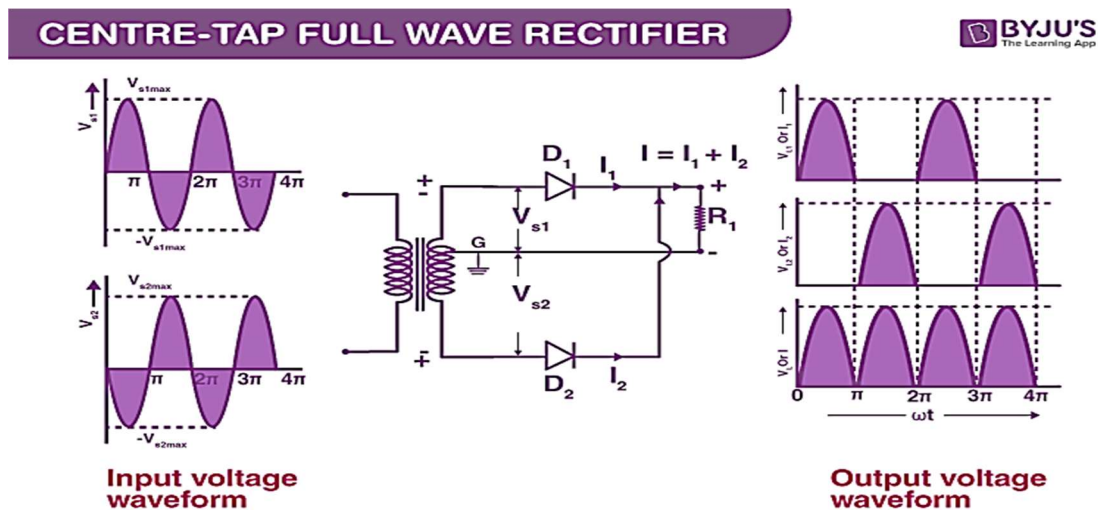
Defining Full Wave Rectifiers

A full wave rectifier is defined as a rectifier that converts the complete cycle of alternating current into pulsating DC.

Unlike halfwave rectifiers that utilize only the halfwave of the input AC cycle, full wave rectifiers utilize the full cycle. The lower efficiency of the half wave rectifier can be overcome by the full wave rectifier.

Full Wave Rectifier Circuit

The circuit of the full wave rectifier can be constructed in two ways. The first method uses a center tapped transformer and two diodes. This arrangement is known as a center tapped full wave rectifier. The second method uses a standard transformer with four diodes arranged as a bridge. This is known as a bridge rectifier. In the next section, we will restrict the discussion to the center tapped full wave rectifier only. You can read our article on bridge rectifier to learn the construction and working of bridge rectifier in detail.



The circuit of the full wave rectifier consists of a step-down transformer and two diodes that are connected and centre tapped. The output voltage is obtained across the connected load resistor.

Working of Full Wave Rectifier

The input AC supplied to the full wave rectifier is very high. The step-down transformer in the rectifier circuit converts the high voltage AC into low voltage AC. The anode of the centre tapped diodes is connected to the transformer's secondary winding and connected to the load resistor. During the positive half cycle of the alternating current, the top half of the secondary winding becomes positive while the second half of the secondary winding becomes negative.

During the positive half cycle, diode D_1 is forward biased as it is connected to the top of the secondary winding while diode D_2 is reverse biased as it is connected to the bottom of the secondary winding. Due to this, diode D_1 will conduct acting as a short circuit and D_2 will not conduct acting as an open circuit

During the negative half cycle, the diode D_1 is reverse biased and the diode D_2 is forward biased because the top half of the secondary circuit becomes negative and the bottom half of the circuit becomes positive. Thus, in a full wave rectifier, DC voltage is obtained for both positive and negative half cycle.

Full Wave Rectifier Formula

Peak Inverse Voltage

Peak inverse voltage is the maximum voltage a diode can withstand in the reverse-biased direction before breakdown. The peak inverse voltage of the full-wave rectifier is double that of a half-wave rectifier. The PIV across D_1 and D_2 is $2V_{max}$.

DC Output Voltage

The following formula gives the average value of the DC output voltage:

$$V_{dc} = I_{av}R_L = \frac{2}{\pi}I_{max}R_L$$

RMS Value of Current

The RMS value of the current can be calculated using the following formula:

$$I_{rms} = \frac{I_{max}}{\sqrt{2}}$$

Form Factor

The form factor of the full wave rectifier is calculated using the formula:

$$K_f = \frac{RMS\ value\ of\ current}{Average\ value\ of\ current} = \frac{I_{rms}}{I_{dc}} = \frac{I_{max}/\sqrt{2}}{2I_{max}/\pi} = \frac{\pi}{2\sqrt{2}} = 1.11$$

Peak Factor

The following formula gives the peak factor of the full wave rectifier:

$$K_p = \frac{Peak\ value\ of\ current}{RMS\ value\ of\ current} = \frac{I_{max}}{I_{max}/\sqrt{2}} = \sqrt{2}$$

Rectification Efficiency

The rectification efficiency of the full-wave rectifier can be obtained using the following formula:

$$\eta = \frac{DC\ Output\ Power}{AC\ Output\ Power}$$

The efficiency of the full wave rectifiers is 81.2%.

Advantages of Full Wave Rectifier

- The rectification efficiency of full wave rectifiers is double that of half wave rectifiers. The efficiency of half wave rectifiers is 40.6% while the rectification efficiency of full wave rectifiers is 81.2%.
- The ripple factor in full wave rectifiers is low hence a simple filter is required. The value of ripple factor in full wave rectifier is 0.482 while in half wave rectifier it is about 1.21.
- The output voltage and the output power obtained in full wave rectifiers are higher than that obtained using half wave rectifiers.